

JEFFREY CHENG

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EDUCATION

Princeton University <i>PhD - Computer Science</i>	Aug 2025 - Present
Johns Hopkins University <i>MS - Computer Science</i>	Aug 2023 - Dec 2024 <i>GPA: 3.85</i>
Duke University <i>BS - Mathematics, Distinction; Computer Science</i>	Aug 2019 - May 2023 <i>GPA: 3.90</i>

EXPERIENCE

Princeton University <i>Research Assistant under Danqi Chen</i> Foundation model research: interpreting silent expert models using chess as a case study, trained a 4B chess-language model to play at Grandmaster strength and explain its reasoning [1]	Aug 2025 - Present <i>Princeton, NJ</i>
Johns Hopkins University <i>Research Assistant under Benjamin Van Durme</i> Foundation model research: identified mismatches between reported and true knowledge cutoffs in open source models [4] and performed pioneering research in latent chain of thought reasoning [3]	Aug 2023 - May 2025 <i>Baltimore, MD</i>
Duke University <i>Research Assistant under Tarek Elgindi</i> Designed candidate time-dependent velocity field to solve open problem concerning total fluid mixing and analytically showed desired properties of candidate velocity field (growth of C^1 norm, decay of H^{-1} norm) [5]	May 2022 - May 2023 <i>Durham, NC</i>
IBM <i>Software Developer Intern</i> - Developed internal and customer facing web applications using ReactJS and Java Spring Boot - Created backend Java API to facilitate functionality and streamline application protection process	May 2022 - Aug. 2022 <i>Chapel Hill, NC</i>
Duke University <i>Research Assistant under Tarek Elgindi</i> Used analytic and numerical methods to analyze potential solutions to the Morrey Conjecture and generated functions satisfying previously proposed theoretical bounds.	May 2021 - Aug 2021 <i>Durham, NC</i>

TEACHING

Johns Hopkins University <i>Teaching Assistant for CS671, Natural Language Processing</i> - Hosted office hours for theoretical and coding homework help, wrote reference solutions for coding homework - Led weekly recitations for 100+ students to explain course content and review practice problems	Aug. 2024 - Dec. 2024
Duke University <i>Teaching Assistant for CS330, Design and Analysis of Algorithms</i> - Hosted office hours for homework help and test preparation in capstone proof-based algorithms class - Contributed to writing homework and exam solutions, graded assignments in weekly grading parties	Dec. 2020 - May. 2023

COMMUNITY

Johns Hopkins Outdoor Programs

Aug. 2023 - Present

Routesetter

- Facilitated safe usage of climbing wall by belaying climbers and inspecting equipment and engaged community by organizing and setting routes for local collegiate competitions

Duke Outdoor Adventures

Aug. 2019 - May 2023

Student Manager and Head Routesetter

- Facilitated safe usage of climbing wall by belaying climbers and inspecting equipment and engaged community by organizing and promoting outdoor trips open to Duke and Durham community
- Led workshops teaching participants climbing techniques and routesetting and managed a team of student routesetters to set for local collegiate competitions

Duke Math Union

Aug 2020 - May 2023

Treasurer

- Wrote problems and hosted Duke Math Meet for hundreds of high school students and collaborated with other members of Duke Putnam Team to teach competition math strategies
- Scored a 26 on the 81st William Lowell Putnam Mathematics Competition

AWARDS

Francis Upton Fellowship, Princeton University

Aug, 2025

Rubenstein Fellowship, Johns Hopkins University

Aug, 2023

SKILLS

Programming: Python, bash, C++, Cython, Javascript, HTML/CSS, R

Frameworks: PyTorch, React.js, Node.js, Express.js, Spring Boot, TensorFlow

Other Tools: Microsoft Office, Git, VSCode, LaTeX

Languages: English, Mandarin (fluent), French (working knowledge)

SELECTED PUBLICATIONS

- [1] A. Bhaskar*, **J. Cheng***, D. Chen, [Language Models that Play Chess and Explain Their Moves](#), *preprint*, 2026.
- [2] W. Jurayj, **J. Cheng**, B. Van Durme, [Is That Your Final Answer? Test-Time Scaling Improves Selective Question Answering](#), in *ACL*, 2025.
- [3] **J. Cheng** & B. Van Durme, [Compressed Chain of Thought: Efficient and Adaptive Reasoning through Dense Representations](#), *preprint*, 2024.
- [4] **J. Cheng**, M. Marone, O. Weller, D. Lawrie, D. Khashabi, B. Van Durme, [Dated Data: Tracing Knowledge Cutoffs in Large Language Models](#), in *COLM*, 2024. **Outstanding Paper Award (0.4%)**
- [5] **J. Cheng** & T. Elgindi, [Mixing and Enhanced Dissipation in Measure Preserving Dynamical Systems](#). Honors thesis, Duke University, 2023.